EE 708: Information Theory and Coding

Quiz II — Solutions to Questions 1 & 2

April 6, 2026

Question 1: Discrete Memoryless Channel $Y = X + Z \pmod{11}$ [4 marks]

Setup. The channel is $Y = X + Z \pmod{11}$, where $X \in \{0, 1, \ldots, 10\}$ and

$$Z = \begin{pmatrix} 1 & 2 & 3 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix},$$

i.e., Z takes values $\{1, 2, 3\}$ each with probability $\frac{1}{3}$, and Z is independent of X.

Part (a): Capacity [2 marks]

The mutual information is

$$I(X;Y) = H(Y) - H(Y \mid X).$$

Step 1: Compute $H(Y \mid X)$.

Given $X = x$, we have $Y = x + Z \pmod{11}$. X and Z are independent wrt each other, so,

$$\begin{aligned} H(Y \mid X = x) &= H(x + Z \quad \text{mod } 11 | X = x) \quad \text{bits} \\ &= H(Z | X = x) \quad \text{bits} \\ &= H(Z) = \log_2 3 \quad \text{bits} \end{aligned}$$

$$H(Y \mid X = x) = H(Z) = \log_2 3 \quad \text{bits},$$

for all x. Therefore,

$$\begin{aligned} H(Y \mid X) &= \sum_x p(x) H(Y \mid X = x) \quad \text{bits}, \\ &= \sum_x p(x) H(Z) \quad \text{bits} \\ &= \log_2 3 \sum_x p(x) \quad \text{bits} \\ &= \log_2 3 \quad \text{bits}. \end{aligned}$$

Step 2: Maximize $H(Y)$.

Since $Y \in \{0, 1, \ldots, 10\}$, the maximum possible entropy is

$$\begin{aligned} &H(Y) \le \log_2(|\mathcal{Y}|) \\ &H(Y) \le \log_2 11 \text{ bits} \end{aligned}$$

with equality achieved when Y is uniformly distributed over $\{0, \dots, 10\}$.

Step 3: Check that uniform Y is achievable.

Suppose X is uniform over $\{0, \dots, 10\}$. Then, you will have Y to be uniform over $\{0, \dots, 10\}$, achieving $H(Y) = \log_2 11$.:

Capacity:

$$\boxed{C = \log_2 11 - \log_2 3 = \log_2\left(\frac{11}{3}\right) \approx 1.874 \text{ bits per channel use.}}$$

Marking Scheme: Part (a) [2 marks]

- **[1 mark]** Computing $H(Y \mid X) = H(Z) = \log_2 3$.
- **[1 mark]** Identifying $C = \log_2 11 - \log_2 3$.

Part (b): Capacity-Achieving Input Distribution $p^*(x)$ [2 marks]

From part (a) we can see that maximum capacity is achieved when Y follows a uniform distribution, which happens when X also follows a uniform distribution over $\{0, 1, \dots, 10\}$. Therefore,

$$\boxed{p^*(x) = \frac{1}{11}, \quad \text{for all } x \in \{0, 1, \dots, 10\}.}$$

Marking Scheme — Part (b) [2 marks]: Stating the correct answer $p^*(x) = \frac{1}{11}$ for all $x \in \{0, \dots, 10\}$. Partial marks will be given in case part (a) is incorrect and valid justification is missing for part (b).

Question 2: Cascading n Copies of a Z-Channel [3 marks]

Setup. The Z-channel has transition probabilities:

$$p(0 \mid 0) = 1, \quad p(1 \mid 0) = 0, \quad p(0 \mid 1) = p, \quad p(1 \mid 1) = 1 - p.$$

That is, input 0 is always received correctly, while input 1 is flipped to 0 with probability p.

The channel transition matrix (rows = input, columns = output) is:

$$T = \begin{pmatrix} 1 & 0 \\ p & 1-p \end{pmatrix}.$$

Cascading Means Series Composition

Cascading n copies means connecting the output of each channel as the input to the next.

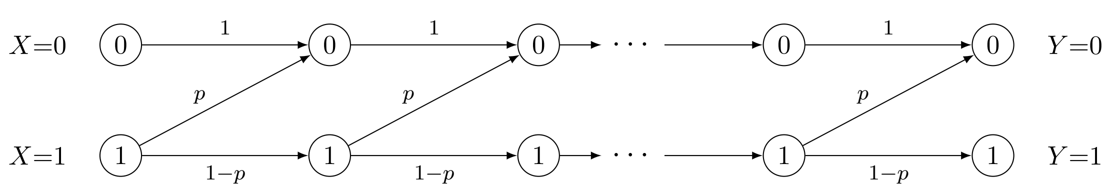


Figure 1: Cascading n Z-channels in series. Input 0 always stays 0; input 1 survives as 1 through each stage with probability $(1 - p)$.

Remarks from the above image

- Input 0 is *never* corrupted at any stage (since $0 \to 0$ always), so $p_{\text{eff}}(0|0) = 1$ is preserved.
- There is no pathway to go from input 0 to output 1, so $p_{\text{eff}}(1|0) = 0$
- Input 1 survives each stage as 1 with probability $(1-p)$; surviving all n stages independently gives $(1-p)^n$. So over n-cascades, $p_{\text{eff}}(1|1) = (1-p)^n$
- With input 1 and output 0, we would have many paths, the effecitve probability of which will be $p_{\text{eff}}(0|1) = 1 - p_{\text{eff}}(1|1) = 1 - (1-p)^n$

Conclusion

The equivalent channel after n cascaded Z-channels has transition probabilities:

$$p_{\text{eff}}(0 \mid 0) = 1, \quad p_{\text{eff}}(1 \mid 0) = 0,$$

$$p_{\text{eff}}(0 \mid 1) = 1 - (1-p)^n, \quad p_{\text{eff}}(1 \mid 1) = (1-p)^n.$$

The equivalent channel is again a Z**-channel** with flip probability

$$p_{\text{eff}} = 1 - (1-p)^n.$$

Marking Scheme — Question 2 [1 + 2 marks]:

- **[1 mark] Nature of the channel:** Identifying that the equivalent channel is again a Z-channel
- **[1 marks] Parameters:** Deriving the effective flip probability $p_{\text{eff}} = 1 - (1-p)^n$
- **[1 marks] Parameters:** Valid justification of the parameters

Q3.

1 mark for correctly stating the dependency of X_i on W and $Y_1, \dots , Y_{i-1}$ for equality (a)

1 mark for correctly stating the requirement of memoryless channel for equality (b) along with a short explanation why (Markov chain created).

1 mark for stating the that feedback-free condition is not required in either statement.

Some contradictory statements were seen where the Markov chain property is mentioned but the memoryless requirement is stated as not required.

Appropriate marks are given depending on the level of correctness in the answers.

Q4.

0.5 marks for calculating $H(Y)$ correctly

0.5 marks for calculating $H(Y|X)$ correctly

1 mark for finding the maximising distribution of X.

Since this was only a 2 mark question, obtaining a simplified answer for Capacity is ignored as long as the maximising distribution is found.

Q3. Equality (a) holds as X_i is a function of W and $Y_1, \ldots, Y_{i-1}$.

X_i is ~~ea~~ formed by encoding the message W and if feedback is used then it also can depend on $Y_1, \ldots, Y_{i-1}$.

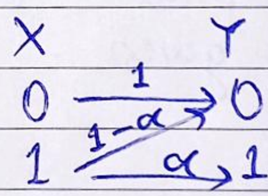
Thus, X_i adds no new information when W and $Y_1, \ldots, Y_{i-1}$ are considered given.

Equality (b) holds because of memorylessness of channel. Current Y_i is independent of previous outputs $Y_1, \ldots, Y_{i-1}$ and only depends on current transmission X_i.

Feedback-free condition is not needed in either equalities while memorylessness is needed in (b).

Q4. $Y = XZ$, $\mathbb{P}(Z=1) = \alpha$, X, Z are independent

This is equivalent to a Z-channel



Let $\mathbb{P}(X=1) = \pi$

By defn; $C = \max_{\pi} H(Y) - H(Y|X)$

Now, $H(Y|X) = p(X=0) \cdot H(Y|X=0) + p(X=1)\, H(Y|X=1)$

$= (1-\pi) \cdot (0) + \pi H(\alpha)$

$= \pi H(\alpha)$

~~H(Y)~~ Now, $p(Y=1) = p(X=1) \cdot p(Z=1)$

$= \pi\alpha$

$\Rightarrow p(Y=0) = 1 - \pi\alpha$

$\therefore H(Y) = H(\pi\alpha)$

Thus, $C = \max\limits_{\pi} H(\pi\alpha) - \pi H(\alpha)$

Let $R = H(\pi\alpha) - \pi H(\alpha)$

Note that $\dfrac{d\,H(x)}{dx} = \dfrac{d}{dx}\left[-x\log x - (1-x)\log(1-x)\right]$

$= -\log x - 1 + \log(1-x) + 1$

$= \log\left(\dfrac{1-x}{x}\right)$

$\therefore \dfrac{dR}{d\pi} = \cancel{\pi}\alpha\log_2\left(\dfrac{1-\pi\alpha}{\pi\alpha}\right) - \cancel{\pi} H(\alpha)$

At maxima, $\dfrac{dR}{d\pi} = 0$

$\Rightarrow \alpha\log_2\left(\dfrac{1-\alpha\pi}{\alpha\pi}\right) = H(\alpha)$

$\Rightarrow \dfrac{1}{\alpha\pi} - 1 = 2^{H(\alpha)/\alpha}$

$\Rightarrow \boxed{\pi = \dfrac{1}{\alpha(\cancel{2}\,1 + 2^{H(\alpha)/\alpha})}}$

$$C = H\left(\frac{1}{1+2^{H(\alpha)/\alpha}}\right) - \frac{H(\alpha)}{\alpha(1+2^{H(\alpha)/\alpha})}$$

$$= \left[\frac{-1}{1+2^{H(\alpha)/\alpha}}\log\left(\frac{1}{1+2^{H(\alpha)/\alpha}}\right)\right] - \left[\left(\frac{2^{H(\alpha)/\alpha}}{1+2^{H(\alpha)/\alpha}}\right)\log\left(\frac{2^{H(\alpha)/\alpha}}{1+2^{H(\alpha)/\alpha}}\right)\right]$$

$$- \left[\frac{H(\alpha)}{\alpha(1+2^{H(\alpha)/\alpha})}\right]$$

3-terms,

the second term can be split as

$$= \frac{2^{H(\alpha)/\alpha}}{1+2^{H(\alpha)/\alpha}} \left(\underbrace{\log\left(2^{H(\alpha)/\alpha}\right)}_{H(\alpha)/\alpha} + \log\left(\frac{1}{1+2^{H(\alpha)/\alpha}}\right)\right)$$

Combining the first term with ~~the se~~ this, we can get

$$= \log\left(\frac{1}{1+2^{H(\alpha)/\alpha}}\right)\left(\frac{-1}{1+2^{H(\alpha)/\alpha}} - \frac{2^{H(\alpha)/\alpha}}{1+2^{H(\alpha)/\alpha}}\right)$$

$$- \frac{H(\alpha)}{\alpha}\left(\frac{2^{H(\alpha)/\alpha}}{1+2^{H(\alpha)/\alpha}} + \frac{1}{1+2^{H(\alpha)/\alpha}}\right)$$

$$= -\log\left(\frac{1}{1+2^{H(\alpha)/\alpha}}\right) - \frac{H(\alpha)}{\alpha}$$

$$= \log\left(1+2^{H(\alpha)/\alpha}\right) - \log\left(2^{H(\alpha)/\alpha}\right)$$

$$= \log\left(1 + 2^{-H(\alpha)/\alpha}\right)$$

(Base of log is 2 everywhere)